test input, full runs — small-fa

L1D 1 sets x 64, L2 1 sets x 512 · metric schema 9 · gem5 TimingSimpleCPU · 4 November 2026

35 of 35 runs completed, 0 running, 0 pending; **0 remaining**.

Program (does not depend on the cache)

run	configs done	spill/ROI	spill ld/load	spill st/store	(spill+csave)/ROI	(spill+csave) ld/load	(spill+csave) st/store
cactuBSSN	1/1	21.32%	51.31%	55.47%	21.42%	51.49%	55.80%
parest	1/1	0.22%	0.56%	1.65%	0.81%	1.48%	12.55%
deepsjeng	1/1	2.78%	9.33%	15.84%	7.31%	22.43%	46.66%
leela	1/1	0.36%	1.25%	1.66%	6.11%	15.91%	44.30%
namd	1/1	3.13%	7.91%	17.27%	3.14%	7.94%	17.37%
lbm	1/1	9.67%	27.79%	35.34%	9.67%	27.79%	35.34%
nab	1/1	7.90%	26.29%	39.86%	8.63%	27.88%	47.43%
xz	1/1	0.68%	2.35%	2.75%	3.30%	9.53%	17.13%
xz_4m_l1	1/1	0.59%	2.08%	2.43%	2.95%	8.51%	15.84%
xz_4m_l2	1/1	0.54%	1.91%	2.32%	2.73%	7.91%	15.62%
xz_4m_l3e	1/1	2.51%	9.02%	25.13%	3.76%	12.59%	42.10%
xz_4m_l4	1/1	2.12%	6.97%	14.43%	4.34%	13.23%	32.30%
xz_4m_l4e	1/1	2.28%	8.15%	25.17%	3.39%	11.30%	42.07%
xz_1m_l0	1/1	0.62%	2.50%	1.68%	3.69%	10.95%	17.49%
xz_1m_l1	1/1	0.58%	2.38%	1.48%	3.48%	10.36%	16.63%
xz_1m_l2	1/1	0.57%	2.35%	1.45%	3.39%	10.13%	16.34%
xz_1m_l3e	1/1	2.59%	9.35%	16.76%	4.60%	15.17%	33.91%
xz_1m_l4	1/1	2.01%	6.65%	9.30%	5.08%	15.11%	26.95%
xz_1m_l4e	1/1	2.43%	8.77%	16.84%	4.27%	14.07%	33.92%
imagick	1/1	0.02%	0.08%	0.05%	4.16%	11.90%	27.17%
blender	1/1	1.16%	2.72%	4.44%	9.37%	20.20%	39.50%
povray	1/1	0.27%	0.45%	0.93%	14.17%	22.21%	55.05%
omnetpp	1/1	0.09%	0.19%	0.36%	12.82%	28.51%	53.40%
xalancbmk	1/1	0.23%	0.53%	0.92%	8.96%	16.83%	45.24%
x264	1/1	6.24%	13.47%	39.69%	8.48%	18.06%	54.69%
gcc	1/1	0.84%	2.18%	3.58%	12.72%	34.38%	52.28%
mcf	1/1	1.58%	2.11%	16.47%	2.74%	3.87%	27.40%
perlbench_makerand	1/1	0.08%	0.00%	0.50%	18.63%	28.89%	59.00%
bwaves_1	1/1	3.55%	9.47%	18.87%	3.55%	9.47%	18.87%
bwaves_2	1/1	3.55%	9.47%	18.87%	3.55%	9.47%	18.87%
exchange2	1/1	5.03%	10.75%	12.58%	6.23%	13.14%	15.81%
fotonik3d	1/1	3.84%	10.81%	9.25%	3.87%	10.86%	9.40%
roms	1/1	4.79%	15.22%	9.67%	4.88%	15.37%	10.29%
wrf	1/1	0.91%	3.41%	2.49%	1.59%	4.94%	7.83%
cam4	1/1	2.11%	7.90%	7.04%	4.52%	14.20%	21.40%
pooled		5.17%	12.33%	27.44%	7.17%	16.47%	39.99%
mean		2.78%	7.88%	12.36%	6.35%	15.89%	31.03%

Miss rates

run	state	L1D	spill ld L1	other ld L1	spill st L1	other st L1	L2	spill ld L2	spill st L2
cactuBSSN	done	15.22%	4.85%	33.94%	5.15%	12.44%	59.94%	12.41%	2.86%
parest	done	9.72%	1.35%	10.44%	1.45%	1.81%	68.60%	3.42%	9.30%
deepsjeng	done	8.08%	1.43%	10.34%	7.23%	4.24%	6.11%	2.44%	11.58%
leela	done	5.07%	0.33%	5.82%	0.42%	3.15%	27.89%	0.04%	98.92%
namd	done	11.61%	2.56%	14.73%	1.18%	2.60%	28.07%	7.56%	16.88%
lbm	done	21.23%	0.00%	7.51%	0.55%	72.77%	99.28%	—	3.25%
nab	done	3.89%	0.13%	6.20%	0.11%	0.75%	8.31%	1.60%	9.58%
xz	done	5.09%	0.01%	6.96%	0.16%	1.73%	31.78%	0.14%	25.60%
xz_4m_l1	done	6.62%	0.01%	8.12%	0.19%	3.93%	41.40%	0.14%	30.88%
xz_4m_l2	done	8.61%	0.01%	10.66%	0.21%	4.63%	47.32%	0.14%	31.02%
xz_4m_l3e	done	6.04%	0.38%	7.38%	0.30%	3.45%	21.48%	0.01%	3.78%
xz_4m_l4	done	6.39%	0.09%	8.04%	0.65%	3.69%	38.80%	0.01%	7.07%
xz_4m_l4e	done	5.90%	0.51%	7.02%	0.32%	3.53%	19.54%	0.01%	3.17%
xz_1m_l0	done	4.62%	0.01%	6.77%	0.49%	0.92%	36.56%	0.46%	11.13%
xz_1m_l1	done	5.20%	0.01%	7.25%	0.54%	1.67%	42.80%	0.46%	11.53%
xz_1m_l2	done	5.49%	0.01%	7.58%	0.55%	1.85%	42.54%	0.46%	11.61%
xz_1m_l3e	done	4.76%	0.22%	6.47%	0.27%	1.67%	23.80%	0.00%	2.67%
xz_1m_l4	done	4.64%	0.02%	6.56%	0.60%	1.64%	34.83%	0.05%	4.10%
xz_1m_l4e	done	4.81%	0.32%	6.37%	0.28%	1.72%	21.91%	0.00%	2.33%
imagick	done	4.16%	7.20%	3.49%	58.93%	5.65%	87.38%	49.88%	64.35%
blender	done	5.13%	2.12%	5.28%	3.87%	5.04%	19.66%	1.62%	1.02%
povray	done	8.70%	7.06%	10.49%	19.52%	4.14%	50.57%	71.81%	68.63%
omnetpp	done	11.53%	0.05%	15.56%	0.77%	4.05%	27.53%	23.39%	1.37%
xalancbmk	done	13.00%	10.87%	16.05%	21.93%	4.63%	34.56%	3.50%	3.19%
x264	done	1.84%	0.16%	2.22%	0.85%	1.91%	31.44%	2.43%	15.18%
gcc	done	5.69%	0.22%	7.26%	1.34%	3.62%	38.30%	15.97%	18.12%
mcf	done	24.52%	3.11%	27.48%	6.86%	9.71%	78.48%	10.04%	17.32%
perlbench_makerand	done	0.08%	0.00%	0.12%	0.26%	0.01%	0.08%	—	0.02%
bwaves_1	done	7.31%	0.86%	7.75%	0.00%	10.68%	95.60%	17.42%	36.61%
bwaves_2	done	7.31%	0.86%	7.75%	0.00%	10.68%	95.60%	17.42%	36.61%
exchange2	done	1.54%	1.69%	2.06%	2.61%	0.64%	1.79%	0.20%	0.70%
fotonik3d	done	8.89%	1.99%	11.55%	0.01%	4.82%	82.63%	1.96%	24.81%
roms	done	10.76%	2.17%	12.36%	0.97%	11.59%	64.31%	10.14%	12.82%
wrf	done	5.88%	5.43%	5.90%	4.49%	5.93%	61.76%	10.42%	19.98%
cam4	done	8.80%	1.46%	8.87%	2.47%	10.55%	59.48%	11.56%	38.18%
pooled		5.59%	1.59%	6.83%	1.94%	4.53%	52.14%	10.51%	8.37%
mean		7.66%	1.64%	9.21%	4.16%	6.34%	43.72%	8.40%	18.75%

Formulas

column	formula	aciklama
spill/ROI	`((spill_load + spill_store) / roi_total_insts) * 100`	ROI komutlarinin yuzde kaci spill load veya spill store
spill ld/load	`(spill_load / roi_load) * 100`	ROI load'larinin yuzde kaci spill reload
spill st/store	`(spill_store / roi_store) * 100`	ROI store'larinin yuzde kaci spill store
(spill+csave)/ROI	`((spill_load + spill_store + csave_load + csave_store) / roi_total_insts) * 100`	ayni oran, callee-save save/restore dahil
(spill+csave) ld/load	`((spill_load + csave_load) / roi_load) * 100`	ROI load'larinin yuzde kaci spill reload veya callee-save restore
(spill+csave) st/store	`((spill_store + csave_store) / roi_store) * 100`	ROI store'larinin yuzde kaci spill store veya callee-save save

column	formula	aciklama
L1D	`(L1D misses / L1D accesses) * 100`	Tum load ve store access'lerinin (spill + other) L1D miss rate'i: L1D'nin bu konfigurasyondaki genel performansi.
spill ld L1	`(spill load L1D misses / spill load L1D accesses) * 100`	Spill load'larin yuzde kaci L1D'de miss oldu. Genelde dusuktur: spill slot'u kisa sure once yazildigi icin cache line cogu zaman hala L1D'dedir.
other ld L1	`((load misses - spill load misses) / (load accesses - spill load accesses)) * 100`	Other load'larin (spill olmayan load'lar) L1D miss rate'i. spill_ld_L1miss ile yan yana bakilir: spill load'lar normal veriden daha mi az, daha mi cok miss oluyor?
spill st L1	`(spill store L1D misses / spill store L1D accesses) * 100`	Spill store'larin yuzde kaci L1D'de miss oldu. Cache write-allocate: miss olan bir store once cache line'i L2'den getirir, sonra yazar.
other st L1	`((store misses - spill store misses) / (store accesses - spill store accesses)) * 100`	Other store'larin (spill olmayan store'lar) L1D miss rate'i; spill_st_L1miss ile karsilastirmak icin.
L2	`(L2 misses / L2 accesses) * 100`	L2'ye gelen tum isteklerin (L1D ve L1I miss'leri) yuzde kaci L2'de de miss oldu ve DRAM'e gitti.
spill ld L2	`(spill load L2 misses / spill load L2 accesses) * 100`	L1D'de miss olan spill load'larin yuzde kaci L2'de de miss oldu. Payda sadece L1D'yi miss eden spill load'lardir, butun spill load'lar degil.
spill st L2	`(spill store L2 misses / spill store L2 accesses) * 100`	L1D'de miss olan spill store'larin yuzde kaci L2'de de miss oldu (payda yine sadece L1D miss'leri).

spill = compiler-confirmed register-allocator spill (GCC `-dP`), csave = callee-saved register save/restore (CFI), both inside the ROI only. pooled = sum of numerators / sum of denominators over the completed runs; mean = average of the per-run percentages. Miss rates are per cache access; an L2 rate's denominator is the L1D misses of that access type.